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Impact of Automation on the Economy

For all of human history, technological advancement has been a cornerstone of what separates us from the rest of the species on the planet. Beginning with our ancient ancestors' creation of primitive stone tools and continuing with the development of agriculture and later the industrial revolution, technology has long impacted our day to day lives by defining how we live in leisure, how we work, and what our work is. Advancements in technology have always come with exciting prospects for the future, while simultaneously causing massive changes in economic structures. Today, the major prospective disruption is coming in the form of Automation and Artificial Intelligence.

Automation and Artificial Intelligence are very exciting fields in technology that are primed for massive advances in the coming future. Like technological advancements of the past, many economists and business owners are very enthusiastic about the prospective increases in efficiency and productivity. However, this will potentially come at a great cost, as many are scared that their jobs will be replaced by automation and artificial intelligence. An example that is often used when discussing the loss of jobs is the taxi, ride sharing, and truck driving industries, which are primed to be impacted heavily with the invention and adoption commercially of self-driving cars. While there is potential for a massive positive impact on the economy from the new technologies, there is also the concern that it could leave many unemployable and stuck in poverty. To illustrate the scope of the concern, it is believed that "54% of EU jobs [are] at risk of computerization." (West, 2018). This means that unless there is an equally large addition of new jobs created by automation and artificial intelligence, as

proponents of automation expect, there will be a large number of workers and employees who will be affected, displaced, or even potentially put out of work entirely.

The major stakeholders in this issue are the Government, the Businesses that are looking to implement automation (and even those who are not), and workers. The government has a vested interest in keeping the economy strong while also making sure unemployment rates do not rise too high. Having large portions of society stuck in unemployment or low to no income status can lead to unrest in the society. However, historically, the US Government has invested very little money in helping citizens deal with economic employment shifts. (Rotman, 2017). Businesses have choices to make in how much to automate. Their main concern is obviously going to be the accumulation of as much profit as possible and can most likely be counted on to automate as much as possible to keep costs low and profits high. Artificial intelligence and automation appear to promise improvements to the economy while also boasting the promise of a higher quality of life overall for consumers. Workers can be divided up into those from low-skill jobs, high-skill jobs, and tech jobs. Lower paying and lower skill jobs are more likely to be replaced by automation. (Rotman, 2017). An example of low-skilled jobs poised to be automated are truck drivers. On the opposite end of the spectrum, an example of high-skilled workers that are poised to benefit from automation and artificial intelligence are software architects and tech workers that are needed to monitor, implement, and develop the automated systems.

Each of the different stakeholder have different claims that they make. Supporters of automation in business make a claim of fact that it will remove the necessity for workers to do low-skilled, repetitive work and allow them to transition into “more valuable, and more fulfilling roles for humans.” (Jesuthasan & Boudreau, 2018). They further state that short term job loss will be offset by new jobs being opened in the future and take the capitalistic approach that the

markets will work themselves out based upon supply and demand. On the other side, those concerned about automation claim of value that robots replacing jobs is a great concern and that steps need to be taken to protect workers. “We need to stop avoiding the situation and create real solutions to help displaced workers.” (Morgan, 2018). Going even further, there are those that make a claim of policy that there need to be steps taken beyond simply providing help to those whose jobs are replaced: “...we need to call automation what it is: a real threat, and a danger to critical human infrastructure.” (Quart, 2017). They argue that automation needs to be slowed down to ensure that workers glean the same benefit from automation as corporations.

As it relates to the future of automation and artificial intelligence, there are a wide range of options that each of the stakeholders can take. The US Government can invest in low income workers likely to be hit hard by automation by providing money for re-education and unemployment protections. The Obama White House outlined three strategies for policy responses to automation and AI growth. The first strategy: “Invest in and develop AI for its many benefits.” The second strategy: “Educate and train Americans for jobs of the future.” The third strategy: “Aid workers in the transition and empower workers to ensure broadly shared growth.” (Executive Office of the President, 2016). Each of these government strategies listed by the Obama White House are aimed towards capitalizing on the advancements of AI without doing anything to slow down the replacement of jobs by automation. It is a tacit approval of the future of automation. The decision making here is governed by a Utilitarian framework ideal of benefitting the majority as much as possible. The choice would be to support automation and deviate from the historical standard of letting the free-market dictate what happens to those displaced by the job loss by investing money in training low-skilled workers to be prepared for the high-skilled tech jobs of the future. An additional government-based solution proposed by

Bill Gates would be to tax the businesses that benefit from automation to create funds for protecting those impacted by automation. (Delaney, 2018). There are pros and cons to government investment. The downside is the extra taxes that people will need to pay to provide the funds for the investments. The upside is the benefit of the investment in providing access to education and training to allow them a higher chance of succeeding in the new automated economy.

Businesses also have options for how they deal with the incoming automation and influx of artificial intelligence tools in their workplaces. Businesses can adopt automation at a level that benefits themselves the most. This is a self-interest approach based on the belief of competition being the best promoter of economic growth. Evidence shows that automation benefits businesses by creating a higher level of profits. (Jesuthasan & Boudreau, 2018). This is supported by the Ego Ethical framework of doing what is best for the company's profits, and the Utilitarian framework idea of automation and artificial intelligence being technology that provides a benefit for most of society, albeit at the expense of low skilled and low-income workers. The pros of this approach are clear: higher profits and growth for the businesses that automate. The cons of this approach are the short-term loss of jobs from low-skilled workers who are unable to find jobs elsewhere.

While probably being the stakeholder with the least amount of agency in the automation debate, workers do have some options that they should be looking into given their economic futures and livelihoods are at stake. Workers can seek out the higher skill jobs that automation and artificial intelligence will create. A somewhat surprising impact of automation is that it leads to a higher level of pay and workplace retention for workers. (Jesuthasan & Boudreau, 2018). As automation and artificial intelligence is adapted by workplaces, typically low level, low skill

work is substituted for higher level, higher skilled work that comes with higher pay, and at longer hours. This is a potential area for workers to capitalize on, and lower workers will need to seek out the education and skills they will need to compete for these positions in the future. This approach has been shown to work. “Students with the right qualifications are able to work with technology instead of being displaced by it...” (Costa, 2017). Alternatively, workers can create organizations or unions demanding that their future be protected by lobbying government and businesses to slow down the growth of automation. (Quart, 2017). This would be motivated by the Care Ethics framework, where they are looking out for the best interests of their families. The options for the worker break down into embracing automation through retraining and education or slowing down automation by opposing it. The pros of the first approach are that workers who succeed in adapting to the new technologies will be rewarded with better work, higher pay, and higher retention. The cons are not every worker is going to be able to succeed in this way, and many workers will not be able to retrain or attain the education that is needed to get the higher paying jobs. As for the second approach, the pros will be workers slowing the automation process and keeping their jobs, while the cons will be the loss of economic and technological progress.

With all the stakeholders and their options laid out, it is now necessary to take a position on what is the best way to move forward. New technology must not be resisted. It must be embraced. There is no doubt that low-skilled and low-paid workers have the highest likelihood of being negatively impacted by the technological advancements that come from automation and artificial intelligence. There is certainly a great concern that many workers that work in specific industries such as truck driving and ridesharing will not have the skills needed to find new jobs once theirs have been replaced by driverless cars. Nevertheless, the benefits of artificial

intelligence and automation to the economy and overall quality of life of most citizens outweighs the negatives of any job loss or job displacement to low skilled workers. This does not mean the US Government and other governments around the world should do nothing to mitigate the damage caused by automation. A broad approach should be taken to emphasize the benefits of increased efficiency, productivity, and improved quality of life for businesses, consumers, and high-skilled tech workers while providing a pathway from low skilled jobs to the jobs of the future. The main goal is to embrace the economy of the future that is fueled by new technologies.

Each of the stakeholders listed have roles to play. Businesses have an obligation to create as much economic growth as possible. Tech companies should not hesitate to create advancements in Artificial Intelligence and automation since that technology will be used to benefit all of society. The government should change course on its historically low level of involvement in programs that help people dealing with changes in job availability. Workers in industries likely to be automated should take the time to prepare as much as possible for the loss of jobs due to automation. This admittedly is a tough proposition, as many likely to be replaced are low skilled workers that will need help to get the skills they need to succeed in the new economy.

The reason it is a superior plan to embrace automation than having workers form organizations to slow the rate of automation is because technology advancements benefit the majority and in the long run lead to a higher standard of living overall. There have been many times in human history where technology changed the way people did business. In each of these cases, it was the prerogative of the worker to adapt skillsets to remain useful in changing economies. The result is a better and more efficient production of goods that benefit society, and often a better quality of life for the worker who now works in a better environment. For example,

the invention of the lightbulb made it difficult for whalers to make money off selling the whale fat used to make candles. Some demand remained, but overall the production of light providing products moved over to electricity based light bulbs, which are common in households and workplaces everywhere. One would be hard-pressed to find someone believing that candle-lit light was better than lightbulbs are today. Another example would be the intercontinental railroads becoming the preferred method for mail transfer over mail delivery by horseback. Certainly, it would require more horseback riders to deliver the same amount of mail in the same amount of time than it would a single train. It would also certainly mean more overall workers employed. That does not change the fact that putting those horseback mail delivery men out of work was part of an overall beneficiary technological advancement to society. This is analogous to the plight of professional drivers today. Just as before, it is to the benefit of society as a whole to embrace the advancements of technology, and the best course of action to take going forward is to embrace automation and artificial intelligence.

Automation and Artificial Intelligence have the potential to heavily impact our economy by displacing many low-skilled and low-income workers. However, technology changes have always been a driving force forward for the quality of life of society in general. There have been many cases already of industries becoming obsolete due to an advancement in technology. By investing in the low skilled workers and finding ways to mitigate the damage of their job loss, society overall can benefit greatly from the promise of automation and artificial intelligence. This is where the government will need to step in and provide investment into our education infrastructure. Having invested in these workers, businesses should confidently proceed with integrating artificial intelligence and automation in as many spaces as possible.

The main difference in this new approach to dealing with employment shifts caused by technology compared to the approaches used in the past is that this approach would also provide investment on the side of the government in providing displaced workers with the skills they need to compete in the new economy. Investment in low skilled workers is a must. There are plenty of tech skills that can be taught. Looking long term, it makes our economy far more competitive to integrate artificial intelligence technology in as many places as possible. Providing workers with the training necessary to work with the new technologies will ensure a robust economy. These workers cannot simply be discarded or left behind. One cannot expect workers to simply be on board with technology changes that would leave many perpetually unemployed. By investing in them, and ensuring they have a pathway to a future working in the new automated, AI integrated economy, the possibility of unrest is diminished or removed entirely. Society is then able to embrace the technology and economy of the future and pave the way for a better tomorrow.

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